

Weak Keys in QC-MDPC-based cryptosystems via the Extended Euclidean Algorithm

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Abstract

In this work we analyze a problem strictly linked with the Rational Reconstruction, which forms the foundation of some post-quantum Quasi-Cyclic Moderate-Density Parity-Check and Quasi-Cyclic Low-Density Parity-Check code-based schemes such as LEDAkem and BIKE. Given a polynomial in a cyclic ring as input, our aim is to recover two polynomials, with specific properties, whose ratio is the input one. The starting point of this work is the paper of Bardet, Dragoi, Luque, and Otmani, which describes some approaches, based on the Extended Euclidean Algorithm, that solves this problem in some specific cases.

In comparison to previous work, we define an additional setting in which the problem can be solved. We also provide an alternative approach to estimate the probability of success, by taking into account a requirement that was not considered in the original paper, thus getting a more precise estimation. Finally, we present a key-recovery attack on BIKE, evaluate its computational cost, and compare it with that of the most efficient known attacks. Although this last step is performed specifically on BIKE, the methodology can be extended to other schemes as well.

1 Introduction

In the 1960s, Gallager [1] introduced the family of Low-Density Parity-Check (LDPC) codes, that have a simple property: they admit a sparse parity-check matrix. This means that the Hamming weight of its row is approximately $\log(n)$, where n is the length of the code. Similar to LDPC codes, Moderate-Density Parity-Check (MDPC)

codes were first introduced in the context of communication [2] as codes admitting a parity-check matrix whose rows have Hamming weight approximately $\sqrt{n}$.

In public-key cryptosystems, LDPC and MPDC codes are often adopted due to their efficient decoding procedures [1, 3], and, to obtain smaller keys, these codes are generally equipped with the Quasi-Cyclic (QC) structure, namely, they admit quasi-circulant sparse parity-check matrices. Since there exists an isomorphism between the set of circulant matrices of order r over the finite field with q elements $\mathbb{F}_q$ and the ring of polynomials $\frac{\mathbb{F}_q[x]}{(x^r-1)}$, a QC-LDPC (resp. QC-MDPC) code can be represented by a sequence of low-density (resp. moderate-density) polynomials. Some of the schemes proposed for the NIST standardization process were based on these types of codes [4, 5].

While these schemes rely on codes with a defined algebraic structure, the most effective known attacks on QC-LDPC and QC-MDPC code-based schemes generally do not focus on this structure, but instead treat the codes as random. *Information Set Decoding* (ISD) is a family of algorithms that represents the most prominent non-structural attack method used against code-based cryptosystems. Starting from Prange’s original ISD algorithm [6], proposed in 1962, numerous variants have since been introduced [7–10], among which one of the most effective is the algorithm by May and Ozerov [11].

In this work, we focus instead on a structural attack that can be run against schemes based on QC-LDPC or QC-MDPC codes, with particular attention to BIKE [4]. The approach underlying this structural attack was first presented in 2016 by Bardet et al. [12]. The authors proposed a method based on the Extended Euclidean Algorithm (EEA) which, by exploiting the relationship between the private and public keys, can recover the private key if it satisfies certain properties. Starting from this approach, they firstly identify some weak keys retrievable in this way and they secondly propose to extend the collection of weak keys using some group actions. In this way they define three sets of weak private keys that can be reconstructed with three methods all based on the EEA. They provide a formula that describes the probability that a private key inside each set is chosen.

Our contribution is the extension of the first set of weak private keys defined in [12], which includes those that can be retrieved with a lower computational cost, thus improving the overall efficiency of the key-recovery attack. We also analyze a variant of the attack, which leads to a second, larger, set of weak keys.

In defining these sets, we introduce an additional condition not mentioned in the original paper, which is, however, necessary for a successful key recovery. Due to this additional study, we obtain a more precise cost estimate of the attacks in [12]. Furthermore, thanks to the extension we identified for the first set of keys, we are able to improve the overall attack cost.

Although our analysis takes as model the BIKE cryptosystem, the methodology can be adapted to other schemes with appropriate modifications.

2 Preliminaries

Throughout this paper, vectors are meant as row vectors unless otherwise specified, i.e. if v is a vector then $v^\top$ is a column vector. Let r be a positive prime integer, and let $\mathcal{R}$ be the cyclic polynomial ring $\frac{\mathbb{F}_2[x]}{(x^r-1)}$. For a polynomial $f(x) = a_0 + a_1x + \dots + a_{r-1}x^{r-1} \in \mathcal{R}$, we denote with $|f|$ the Hamming weight of the vector $(a_0, a_1, \dots, a_{r-1}) \in \mathbb{F}_2^r$ composed of the coefficients of $f(x)$. Given a set A , we use the notation $a \xleftarrow{\$} A$, to indicate that a is chosen from A following a uniform distribution.

The problem that we are interested in is a variation of the Rational Reconstruction Problem, formally stated in Problem 1.

Problem 1 *Let h_0, h_1 be two polynomials in $\mathcal{R}$, such that $|h_0| = w_0$, $|h_1| = w_1$, with w_0, w_1 odd integers such that $1 < w_0 + w_1 < r$. Let $h = h_1 h_0^{-1} \in \mathcal{R}$. On input h determine h_0 and h_1 .*

We remark that Problem 1 might have multiple valid solutions, i.e. there might also exist $\bar{h}_0 \neq h_0, \bar{h}_1 \neq h_1 \in \mathcal{R}$ with $\bar{h}_0, \bar{h}_1 \neq 0$ such that $|\bar{h}_0| = w_0$, $|\bar{h}_1| = w_1$ with $h = \bar{h}_1 \bar{h}_0^{-1} \in \mathcal{R}$. Whenever we are given an instance of Problem 1 where h is known to have a solution (h_0, h_1) we are thus interested in finding any solution $(\bar{h}_0, \bar{h}_1)$ regardless of it being equal to (h_0, h_1) .

In some specific cases Problem 1 can be solved using a method based on the Extended Euclidean Algorithm (described in Algorithm 1). On input $(x^r - 1, h)$, at each step the EEA will generate a tuple s_i, r_i such that

$$s_i h \equiv r_i \pmod{(x^r - 1)}.$$

When the polynomials h_0 and h_1 satisfy certain properties, Problem 1 can be efficiently solved by simply checking whether the partial output s_i, r_i is a valid solution for Problem 1. The main difficult task is to define and estimate precisely the number of polynomials h_0, h_1 that can be recovered through the EEA.

In [12], the authors focus on specific cases in which they claim that Problem 1 can be solved. They identify three main subsets of

$$\mathcal{P} := \{(h_0, h_1) \in \mathcal{R} \mid |h_i| = w_i, \text{ and } w_i \text{ odd}\},$$

for which, whenever a solution (h_0, h_1) of an instance h of Problem 1 belong to them, then the instance can be efficiently solved by means of the EEA.

We present now the three cases identified in [12].

Case i.

The first subset $W_1 \subset \mathcal{P}$ is defined as

$$W_1 := \{(h_0, h_1) \in \mathcal{P} \mid \deg(h_0) + \deg(h_1) < r\}.$$

Algorithm 1: Extended Euclidean Algorithm (EEA) over $\mathbb{F}_2[x]$

Input: Two nonzero polynomials a, b in $\mathbb{F}_2[x]$ such that $\deg(a) \geq \deg(b)$
Output: The gcd (a, b)

```
1  $(r_0, r_1) \leftarrow (a, b), (s_0, t_0) \leftarrow (1, 0), (s_1, t_1) \leftarrow (0, 1);$ 
2  $i \leftarrow 1;$ 
3 while  $r_i \neq 0$  do
4    $q_i \leftarrow r_{i-1} \text{ div } r_i;$  // div = integer division
5    $r_{i+1} \leftarrow r_{i-1} - q_i r_i;$ 
6    $s_{i+1} \leftarrow s_{i-1} - q_i s_i;$ 
7    $t_{i+1} \leftarrow t_{i-1} - q_i t_i;$ 
8    $i \leftarrow i + 1;$ 
9  $\nu \leftarrow i - 1;$ 
10 return  $r_\nu$ 
```

For W_1 they provide a precise formula that describes its cardinality (Equation 12 in [12]) and the probability that a solution (h_0, h_1) of a random instance h lies inside this set.

Case ii.

The second subset $W_2 \subset \mathcal{P}$ is defined as

$$W_2 := \{(x^i h_0, x^j h_1) \in \mathcal{P} \mid i, j \in \mathbb{Z}_r, \deg(h_0) + \deg(h_1) < r\}.$$

Also in this case they provide a precise formula (Equation 19 in [12]) for the probability that a solution to a random instance h lies inside W_2 .

Case iii.

Let α be an element of $\mathbb{Z}_r^*$, and let $f = \sum_{i=0}^{r-1} a_i x^i$ be a polynomial in $\mathcal{R}$. Let us define the action $\cdot$ of $\mathbb{Z}_r^*$ on $\mathcal{R}$ by the rule

$$\alpha \cdot f := \sum_{i=0}^{r-1} a_i x^{i \cdot \alpha}.$$

The last subset, $W_3 \subset \mathcal{P}$, is

$$W_3 := \{(\alpha \cdot x^i h_0, \alpha \cdot x^j h_1) \in \mathcal{P} \mid i, j \in \mathbb{Z}_r, \alpha \in \mathbb{Z}_r^*, \deg(h_0) + \deg(h_1) < r\}.$$

In this last case the authors provide an upper and lower bound (Equation 26 in [12]) for the probability of randomly selecting an instance whose solution is in W_3 .

We remark that the cost of solving Problem 1 if the input polynomial h has a solution in W_1 differs from the other two cases. Specifically, the cost associated with elements in W_1 is lower than for the second case W_2 , which in turn is lower than the third case W_3 .

In this work we identify new sets for which Problem 1, in the variation with $w_0 = w_1$, can be efficiently solved. The following useful and well-known theorem, whose variants are often of central importance in decoding algorithm, see e.g. [13], represents the foundation of the approach.

Theorem 1 *Let a and b be two nonzero polynomials in $\mathbb{F}_2[x]$ such that $\deg(a) \geq \deg(b)$. Let t and r be two nonzero polynomials in $\mathbb{F}_2[x]$ such that:*

1. $\gcd(r, t) = 1$;
2. $\deg(t) + \deg(r) < \deg(a)$;
3. $tb \equiv r \pmod{a}$.

Then, by computing the Extended Euclidean Algorithm, there exists an index $h \in \{0, \dots, \nu\}$ such that $t = t_h$ and $r = r_h$.

From now on, whenever we talk about Problem 1, we always consider instances for which $w_0 = w_1 = d$, namely, the two polynomials h_0 and h_1 have d nonzero coefficients.

3 Characterization of some solvable cases

We now describe the setting in which we aim to solve Problem 1, and identify specific cases that can be solved using methods based on the EEA.

Definition 1 A *weak polynomial pair* is a pair of polynomials (h_0, h_1) such that Problem 1 can be solved when given $h := h_1 h_0^{-1}$ as input.

Let h_0 and h_1 be two polynomials in $\mathcal{R}$ such that $|h_0| = |h_1| = d$, with d an odd integer, and let $h = h_0^{-1} h_1$. Note that, in this setting, h_0 and h_1 are both invertible in $\mathcal{R}$, and $h \in \mathcal{R}$ is well-defined. If we consider the polynomials in $\mathbb{F}_2[x]$, then there exists a polynomial $q \in \mathbb{F}_2[x]$ such that

$$q(x^r - 1) + h_0 h = h_1.$$

Our goal is to solve Problem 1 on input h obtained as above, namely, to determine either the polynomials h_0 and h_1 or two alternative polynomials $\bar{h}_0$ and $\bar{h}_1$ in $\mathcal{R}$, such that $|\bar{h}_0| = |\bar{h}_1| = d$ and $h = \bar{h}_0^{-1} \bar{h}_1$ in $\mathcal{R}$. To achieve our purpose we consider two cases in which the polynomials h_0 and h_1 have specific properties that can be exploited to successfully apply Theorem 1.

Case 1:

There exist an $i \in \mathbb{Z}_r$ and two polynomials $\tilde{h}_0$ and $\tilde{h}_1$, that are coprime in $\mathbb{F}_2[x]$ and satisfy $\deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r$, such that

$$(h_0, h_1) = (x^i \tilde{h}_0 \mod (x^r - 1), x^i \tilde{h}_1 \mod (x^r - 1)).$$

In this scenario, the shift x^i applied to $\tilde{h}_0$ and $\tilde{h}_1$ disappears when h is computed, indeed,

$$h \equiv h_0^{-1} h_1 \mod (x^r - 1) \equiv (x^i \tilde{h}_0)^{-1} x^i \tilde{h}_1 \mod (x^r - 1) \equiv \tilde{h}_0^{-1} \tilde{h}_1 \mod (x^r - 1).$$

Consequently, a solution can be found by applying the Modified Extended Euclidean Algorithm (MEEA) presented in Algorithm 2 on input $(x^r - 1), h$ and d . Let us call K_1 the set of pairs of polynomials with these properties, i.e.

$$K_1 := \{(x^i \tilde{h}_0 \mod (x^r - 1), x^i \tilde{h}_1 \mod (x^r - 1)) \in \mathbb{F}_2[x]^2 \mid i \in \mathbb{Z}_r, \gcd(\tilde{h}_0, \tilde{h}_1) = 1, \deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r, |\tilde{h}_0| = |\tilde{h}_1| = d\}. \quad (1)$$

Notice that in [12], the authors define Case i. by working with the set W_1 containing pairs of polynomials for which Problem 1 can be solved by applying the MEEA. W_1 is approximately the subset of K_1 corresponding to the sole case $i = 0$. More precisely in K_1 there is the comprimality constraint, that is not consider in W_1 . Keeping in mind this fact we can think of Case 1. as a generalization of Case i.

Algorithm 2: Modified Extended Euclidean Algorithm (MEEA) over $\mathbb{F}_2[x]$

Input: Two nonzero polynomials a, b in $\mathbb{F}_2[x]$ such that $\deg(a) \geq \deg(b)$ and an integer $d > 0$

Output: Two polynomials (r_i, t_i) with $|r_i|, |t_i| = d$, such that $r_i \equiv t_i b \mod a$, or **failure** if the algorithm does not find these polynomials

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1  $(r_0, r_1) \leftarrow (a, b), (s_0, t_0) \leftarrow (1, 0), (s_1, t_1) \leftarrow (0, 1);$ 
2  $i \leftarrow 1;$ 
3 while  $r_i \neq 0$  do
4   if  $|r_i|, |t_i| = d$  then
5     return  $(r_i, t_i);$ 
6    $q_i \leftarrow r_{i-1} \text{ div } r_i;$ 
7    $r_{i+1} \leftarrow r_{i-1} - q_i r_i;$ 
8    $s_{i+1} \leftarrow s_{i-1} - q_i s_i;$ 
9    $t_{i+1} \leftarrow t_{i-1} - q_i t_i;$ 
10   $i \leftarrow i + 1;$ 
11 return failure
```

Case 2:

There exist two integers $i, j \in \mathbb{Z}_r$ and two polynomials $\tilde{h}_0, \tilde{h}_1$ coprime in $\mathbb{F}_2[x]$ with $\deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r$, such that

$$(h_0, h_1) = (x^i \tilde{h}_0 \mod (x^r - 1), x^j \tilde{h}_1 \mod (x^r - 1)).$$

In this case the shifts of the two polynomials are not directly visible in h , but their difference remains. More precisely

$$\begin{aligned} h &\equiv h_0^{-1} h_1 \mod (x^r - 1) \equiv (x^i \tilde{h}_0)^{-1} x^j \tilde{h}_1 \mod (x^r - 1) \equiv x^{j-i} \tilde{h}_0^{-1} \tilde{h}_1 \mod (x^r - 1) \\ &\implies x^{i-j} h \equiv \tilde{h}_0^{-1} \tilde{h}_1 \mod (x^r - 1). \end{aligned}$$

Consequently, $\tilde{h}_0$ and $\tilde{h}_1$ can be found by applying the Modified Extended Euclidean Algorithm on each shift of h . Once an output different from **failure** is given, what we get are two polynomials $\tilde{h}_0$ and $\tilde{h}_1$ of weight d such that, for a certain k ,

$$\tilde{h}_1 \equiv \tilde{h}_0 x^k h \mod (x^r - 1).$$

At this point, by defining $\bar{h}_0 := \tilde{h}_0 x^k \mod (x^r - 1)$ and $\bar{h}_1 := \tilde{h}_1 \mod (x^r - 1)$, we obtain

$$h \equiv \bar{h}_0^{-1} \bar{h}_1 \mod (x^r - 1), \quad \text{with } |\bar{h}_1| = |\bar{h}_2| = d.$$

The pairs that fall into this case are thus given by the following set

$$\begin{aligned} K_2 := \{ &(x^i \tilde{h}_0 \mod (x^r - 1), x^j \tilde{h}_1 \mod (x^r - 1)) \in \mathbb{F}_2[x]^2 \mid i, j \in \mathbb{Z}_r, \\ &\gcd(\tilde{h}_0, \tilde{h}_1) = 1, \deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r, |\tilde{h}_0| = |\tilde{h}_1| = d\}. \end{aligned} \quad (2)$$

It is easy to note that $K_1 \subset K_2$. However, unlike the set K_1 , whose elements can be found by using Algorithm 2 on input $(x^r - 1, h, d)$, the elements of K_2 require the application of the same algorithm several times on different shifts of h . The latter is presented in detail in Algorithm 3.

Remark 1 From a theoretical point of view, the difference between $\mathcal{W}_2$ and K_2 lies in the requirement that h_0 and h_1 are coprime, so that Theorem 2 can be successfully applied and thus that the solution to Problem 1 can be correctly determined by applied the MEEA. Note that, without the coprimality assumption, the algorithm fails to determine h_0 and h_1 , and this fact leads to an overestimation of the probability (Equation 19 of [12]) of selecting instances that are solvable and consequently to an underestimation of the cost of the attack.

To summarize, we have just described two sets for which Problem 1 can be solved. If $(h_0, h_1) \in K_1$, then the solution can be found by running Algorithm 2 on input $(x^r - 1, h, d)$. Otherwise, if $(h_0, h_1) \in K_2$ Algorithm 3 has to be run on the same input $(x^r - 1, h, d)$.

Algorithm 2 solves the problem for a smaller subset of polynomials, nevertheless, as we

Algorithm 3: MEEA over $\mathbb{F}_2[x]$ for different shifts

Input: Two nonzero polynomials a, b in $\mathbb{F}_2[x]$ such that $\deg(a) \geq \deg(b)$ and an integer $d > 0$

Output: Two polynomials (r_i, t_i) and an integer $k \in \{0 \dots r-1\}$ such that $|r_i| = |t_i| = d$ and $r_i \equiv t_i b x^k \pmod{a}$, or **failure** if the algorithm does not find such (r_i, t_i) and k

```

1 for  $i \leftarrow 0$  to  $r-1$  do
2    $b_{\text{shifted}} \leftarrow x^i b \pmod{a}$  ;
3    $\text{res} \leftarrow \text{Algorithm2}(a, b_{\text{shifted}}, d)$  ;
4   if  $\text{res} \neq \text{failure}$  then
5     return  $(\text{res}, i)$ ;
6 return failure

```

will see in Section 4, it has a lower computational cost with respect to Algorithm 3. In contrast, the latter is effective for a wider range of polynomials but is more expensive.

4 Complexity analysis

Both Algorithm 2 and 3 rely on the EEA. Let a and b be the two polynomials in $\mathbb{F}_2[x]$ given as input to the EEA, with $\deg(a) > \deg(b)$. At every step of the algorithm the most expensive operation computed is the division between the two polynomials r_i and r_{i+1} . Since $\deg(r_{i+1}) < \deg(r_i)$, the time complexity of each step can be estimated as $\mathcal{O}(\deg(r_i)(\deg(r_i) - \deg(r_{i+1})))$. Indeed the division algorithm performs $\deg(r_i) - \deg(r_{i+1})$ iterations and each iteration involves approximately $\deg(r_i)$ arithmetic operations.

Thus, the overall complexity $\mathbf{c}_{\text{EEA}}$ of the algorithm is obtained by multiplying the complexity per step by the total number of steps. More formally

$$\begin{aligned}
\mathbf{c}_{\text{EEA}} &= \mathcal{O}(\deg(r_0)(\deg(r_0) - \deg(r_1)) + \deg(r_1)(\deg(r_1) - \deg(r_2)) + \dots \\
&\quad + \deg(r_{\nu-1})(\deg(r_{\nu-1}) - \deg(r_\nu))) \\
&\approx \mathcal{O}(\deg(r_0)(\deg(r_0) - \deg(r_1)) + \deg(r_0)(\deg(r_1) - \deg(r_2)) + \dots \\
&\quad + \deg(r_0)(\deg(r_{\nu-1}) - \deg(r_\nu))) \\
&= \mathcal{O}(\deg(r_0)[(\deg(r_0) - \deg(r_1)) + (\deg(r_1) - \deg(r_2)) + \dots \\
&\quad + (\deg(r_{\nu-1}) - \deg(r_\nu))]) \\
&= \mathcal{O}(\deg(r_0)[\deg(r_0) - \deg(r_\nu)]) \\
&\approx \mathcal{O}((\deg r_0)^2) = \mathcal{O}((\deg a)^2).
\end{aligned}$$

We remark that the difference between the EEA and Algorithm 2 is the adoption of a new stopping condition. In particular 2 is run with input a polynomial a of degree

r and a polynomial b of degree smaller than r . Therefore, the complexity is

$$\mathbf{c}_{\text{Algorithm 2}} = \mathcal{O}((\deg a)^2) = \mathcal{O}(r^2).$$

In the same way, Algorithm 3 requires in our scenario at most r iterations of Algorithm 2. Consequently, its complexity is

$$\mathbf{c}_{\text{Algorithm 3}} = \mathcal{O}(r(\deg a)^2) = \mathcal{O}(r^3).$$

5 Probability of selecting weak polynomials

In this section we study the probability of selecting a pair of polynomials (h_0, h_1) for which Problem 1 is solvable via the two described approaches.

Our strategy involves, as first step, calculating the cardinalities of the sets K_1 and K_2 , defined respectively in Equation (1) and Equation (2). Secondly, computing the ratio between these cardinalities and the total number of possible pairs of polynomials for Problem 1.

In order to calculate the cardinality of K_1 and K_2 , we proceed as follows. First, we relax the coprimality condition used to define these sets and we define in this way two modified sets: $\mathcal{W}_1$ and $\mathcal{W}_2$.

$$\mathcal{W}_1 := \{(x^i \tilde{h}_0, x^i \tilde{h}_1) \in \mathcal{R}^2 \mid \deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r, |\tilde{h}_0| = |\tilde{h}_1| = d, i \in \mathbb{Z}_r\}, \quad (3)$$

$$\mathcal{W}_2 := \{(x^i \tilde{h}_0, x^j \tilde{h}_1) \in \mathcal{R}^2 \mid \deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r, |\tilde{h}_0| = |\tilde{h}_1| = d, i, j \in \mathbb{Z}_r\}. \quad (4)$$

Then, we compute the cardinality of the modified sets by using combinatorial methods. In a second phase, in order to estimate the cardinality of the original sets, we perform a statistical analysis to determine how many pairs within the modified sets satisfy the relaxed coprimality condition.

First of all, let us consider the set of all possible polynomials pairs. Let d be an odd integer such that $1 < d < r$, then

$$\mathcal{P}_d := \{(h_0, h_1) \in \mathcal{R} \mid |h_0| = |h_1| = d\},$$

hence,

$$|\mathcal{P}_d| = \binom{r}{d}^2.$$

The probability of selecting a pair of polynomials (h_0, h_1) , for which we can solve Problem 1 through the first and second approach is, respectively,

$$\mathbb{P}_1 := \frac{|K_1|}{|\mathcal{P}_d|}, \quad \mathbb{P}_2 := \frac{|K_2|}{|\mathcal{P}_d|}. \quad (5)$$

Before entering the details of the analysis, let us state an important theorem, presented in [14], that will be needed later and a definition.

Theorem 2 Let $n, k, l \in \mathbb{N}^+$. The number of ways to split n balls into l bins such that the maximum number of balls inside each bin is at most k , is equal to

$$R(n, l, k) := \sum_{t=0}^l (-1)^t \binom{l}{t} \binom{n - t(k+1) + l - 1}{l - 1}.$$

Definition 2 Consider a register. A *run of ones* is a sequence of consecutive ones both preceded and followed by a zero, while a *run of zeros* is a sequence of consecutive zeros preceded and followed by a one.

5.1 Cardinality of the sets $\mathcal{W}_1$ and $\mathcal{W}_2$

Let us start from case 2, followed by case 1.

5.1.1 Case 2:

We first introduce some lemmas that will be used in the proof of the main results.

Lemma 1 Let r and d be two positive integers with $d < r$. The number of ways in which d ones and $r - d$ zeros can be distributed into a circular register of length r with a fixed starting position, given by $\binom{r}{d}$, can be rewritten as

$$r \sum_{v=1}^d \sum_{\substack{j_1, \dots, j_v \\ j_1 + \dots + j_v = r - d}} \binom{d-1}{v-1} \frac{1}{v},$$

or equivalently

$$r \sum_{v=1}^d \sum_{\substack{w_1, \dots, w_v \\ w_1 + \dots + w_v = r - d - v}} \binom{d-1}{v-1} \frac{1}{v},$$

where v represents the number of runs of ones, and $(j_1, \dots, j_v)$ represents the length of the zero runs.

Proof Consider a circular register of length r , with a fixed starting position, and suppose to have d ones and $r - d$ zeros to place inside this register. For now we place a one in the first position of the register and a zero in the last. In a second phase we will consider all the shifts of every configuration.

Let us focus on the possible number of runs of ones. Since there are a total of d ones, the number of runs is between 1 and d . Note that, the starting point of each run is characterized by a one preceded by a zero. Let v in $\{1, \dots, d\}$ be the number of runs of ones. Note that, among these v ones preceded by a zero, we also count the one in the first position as being preceded by the zero in the last position (since we work with a circular register). Thus, we are left with $d - v$ ones to distribute among the runs. The number of ways to do this can be easily determined using the stars and bars method, and it is

$$\binom{(d-v) + (v-1)}{v-1} = \binom{d-1}{v-1}.$$

To avoid counting more times the same configuration, when considering the shift, we divide the just computed value by v . So, the remaining $d - v$ ones could be distributed in

$$\binom{d-1}{v-1} \frac{1}{v}$$

ways.

At this point, we can focus on how distribute the zeros. For a fixed number v of runs of ones, there are also v runs of zeros. We are interested not only in the number of ways to distribute the $r - d - v$ remaining zeros, but we also want to know the length of each run. Let the tuple $(j_1, \dots, j_v)$ represent the length of the v runs of zero, then the following sum describes the number of ways in which the $r - d - v$ zeros could be distributed among the runs.

$$\sum_{\substack{j_1, \dots, j_v \mid j_i > 0 \ \forall i \\ j_1 + \dots + j_v = r - d}} 1 = \sum_{\substack{w_1, \dots, w_v \mid \\ w_1 + \dots + w_v = r - d - v}} 1.$$

Summarizing, with the constraint that in the first position of the register there is a one, in the last position there is a zero, and without counting the shifts, the number of ways in which d ones and $r - d$ zeros could be distributed over a register of length r is

$$\sum_{v=1}^d \sum_{\substack{j_1, \dots, j_v \mid j_i > 0 \ \forall i \\ j_1 + \dots + j_v = r - d}} \binom{d-1}{v-1} \frac{1}{v}$$

or, equivalently,

$$\sum_{v=1}^d \sum_{\substack{w_1, \dots, w_v \mid \\ w_1 + \dots + w_v = r - d - v}} \binom{d-1}{v-1} \frac{1}{v}.$$

At this stage, we can consider the r possible shifts, thereby obtaining the formula stated in the lemma. $\square$

Lemma 2 *Let r, d and m be positive integers such that $m, d < r$ and consider the following set*

$$H := \{(j_1, \dots, j_v) \mid \sum_{i=1}^v j_i = r - d, \ 1 \leq j_i \leq m, \ \exists k : j_k = m\}.$$

Then,

$$|H| = R(r - d - v, v, m - 1) - R(r - d - v, v, m - 2).$$

Proof First of all, note that the cardinality of H is equal to the cardinality of the set $\tilde{H}$, defined as

$$\tilde{H} := \{(j_1, \dots, j_v) \mid \sum_{i=1}^v j_i = r - d - v, \ 0 \leq j_i \leq m - 1, \ \exists k : j_k = m - 1\}.$$

$\tilde{H}$ represents the ways in which $r - d - v$ balls can be stored into v bins with capacity $m - 1$, with at least one bin that contains $m - 1$ balls. Its cardinality can be seen as the number of ways in which $r - d - v$ balls can be stored into v bins with capacity $m - 1$, minus the number of ways in which $r - d - v$ balls can be stored into v bins with capacity $m - 2$. From

Theorem 2, it can be computed by the formula

$$R(r-d-v, v, m-1) - R(r-d-v, v, m-2).$$

□

Lemma 3 *Let r, d and m be positive integers such that $m, d < r$ and consider the following set*

$$T = \left\{ (q_1, \dots, q_k) \mid \sum_{i=1}^k q_i = r-d, \ q_i > 0 \ \forall i, \ \max\{q_i \mid i \in \{1, \dots, k\}\} \geq r-(m+1) \right\}.$$

Then

$$|T| = \binom{r-d-1}{k-1} - R(r-d-k, k, r-m-3).$$

Proof First, note that the cardinality of T is equal to the cardinality of the following set

$$\bar{T} = \left\{ (q_1, \dots, q_k) \mid \sum_{i=1}^k q_i = r-d-k, \ \max\{q_i \mid i \in \{1, \dots, k\}\} \geq r-(m+1)-1 \right\}.$$

At this point the set $\bar{T}$ can be written as the difference between two other sets, more formally as

$$\left\{ (q_1, \dots, q_k) \mid \sum_{i=1}^k q_i = r-d-k \right\} \setminus \left\{ (q_1, \dots, q_k) \mid \sum_{i=1}^k q_i = r-d-k, \ 0 \leq q_i \leq r-(m+1)-2 \ \forall i \right\}.$$

Hence, the searched cardinality can be thought of as the number of ways of distributing $r-d-k$ balls inside k bins minus the number of ways of distributing $r-d-k$ balls inside k bins with capacity $r-m-3$. Consequently, by Theorem 2 together with the stars and bars method, we get that

$$|T| = \binom{r-d-1}{k-1} - R(r-d-k, k, r-m-3).$$

□

To study the cardinality of K_2 , let us first consider $\mathcal{W}_2$, defined in Equation (4), namely, the superset of K_2 where the requirement of coprimality is removed.

Theorem 3 *Let*

$$\mathcal{W}_2 := \{(x^i \tilde{h}_0, x^j \tilde{h}_1) \in \mathcal{R}^2 \mid \deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r, \ |\tilde{h}_0| = |\tilde{h}_1| = d, \ i, j \in \mathbb{Z}_r\}.$$

Then,

$$|\mathcal{W}_2| = r^2 \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v} \cdot \\ \left(\sum_{k=1}^d \left(\binom{r-d-1}{k-1} - R(r-d-k, k, r-m-3) \right) \binom{d-1}{k-1} \frac{1}{k} \right).$$

Proof To compute $|\mathcal{W}_2|$, firstly, we focus on the polynomial $x^i \tilde{h}_0$, that can be seen as the distribution of d ones and $r-d$ zeros over a register of length r . For Lemma 1, it can be constructed in

$$r \sum_{v=1}^d \sum_{\substack{j_1, \dots, j_v | j_i > 0 \ \forall i \\ j_1 + \dots + j_v = r-d}} \binom{d-1}{v-1} \frac{1}{v}$$

different ways. Now, for every polynomial $x^i \tilde{h}_0$ we have to count the number of polynomials $x^j \tilde{h}_1$ such that $(x^i \tilde{h}_0, x^j \tilde{h}_1) \in \mathcal{W}_2$.

Let v be the number of runs of ones and zeros of $x^i \tilde{h}_0$ and let $(j_1, \dots, j_v)$ be the tuple that describes the zeros runs. We want to count the number of polynomials that can be written as $x^j \tilde{h}_1 \in \mathcal{R}$, with $\deg(\tilde{h}_1) \leq \max\{j_i \mid i \in \{1, \dots, v\}\}$.

Let m be the $\max\{j_i \mid i \in \{1, \dots, v\}\}$. The requirement that $\tilde{h}_1$ has to have degree at most m , can be translated into the request that there has to be one run of zeros with at least $r - (m+1)$ elements.

Hence, using Lemma 1, with the latter constrain, we get that the number of ways in which $x^j \tilde{h}_1$ can be chosen is

$$r \sum_{k=1}^d \sum_{\substack{q_1, \dots, q_k | q_i > 0 \ \forall i \\ q_1 + \dots + q_k = r-d, \\ \max\{q_i\} \geq r-(m+1)}} \binom{d-1}{k-1} \frac{1}{k},$$

where k describes the number of runs of one and $(q_1, \dots, q_k)$ describes the length of the runs of zeros.

Consequently,

$$|\mathcal{W}_2| = r \sum_{v=1}^d \sum_{\substack{j_1, \dots, j_v | j_i > 0 \ \forall i \\ j_1 + \dots + j_v = r-d}} \binom{d-1}{v-1} \frac{1}{v} \left(r \sum_{k=1}^d \sum_{\substack{q_1, \dots, q_k | q_i > 0 \ \forall i \\ q_1 + \dots + q_k = r-d, \\ \max\{q_i\} \geq r-(m+1)}} \binom{d-1}{k-1} \frac{1}{k} \right).$$

Although we have derived a formula, this is not yet sufficient. Let us now refine it in order to obtain a more computable version, as in the theorem statement.

Note that, in the summation over the tuples $(j_1, \dots, j_v)$, the only relevant information to retain is the number of tuples that have the same maximum value m and which is this value. Observe that m can range between $\lceil \frac{r-d}{v} \rceil$ and $r-d-(v-1)$.

Due to Lemma 2, the cardinality of $\mathcal{W}_2$ can be rewritten as

$$|\mathcal{W}_2| = r \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v}.$$

$$r \sum_{k=1}^d \sum_{\substack{q_1, \dots, q_k \mid q_i > 0 \ \forall i \\ q_1 + \dots + q_k = r-d, \\ \max\{q_i\} \geq r-(m+1)}} \binom{d-1}{k-1} \frac{1}{k}.$$

Due to Lemma 3, we can enhance the latter equation in the following way

$$|\mathcal{W}_2| = r^2 \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v}.$$

$$\left(\sum_{k=1}^d \left(\binom{r-d-1}{k-1} - R(r-d-k, k, r-m-3) \right) \binom{d-1}{k-1} \frac{1}{k} \right).$$

□

5.1.2 Case 1:

Let us now focus on $\mathcal{W}_1$, described by Equation (3), that denote the set of polynomials pairs corresponding to case 1, where the coprimality requirement is relaxed.

Theorem 4 *Let*

$$\mathcal{W}_1 := \{(x^i \tilde{h}_0, x^i \tilde{h}_1) \in \mathcal{R}^2 \mid \deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r, |\tilde{h}_0| = |\tilde{h}_1| = d, i \in \mathbb{Z}_r\}.$$

Then,

1. *the cardinality of $\mathcal{W}_1$ is bounded below by*

$$r \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v} \binom{m+1}{d};$$

2. *the cardinality of $\mathcal{W}_1$ is bounded above by*

$$r \sum_{s=2d-2}^{r-1} \sum_{d_1=d-1}^{s-(d-1)} \left[\binom{d_1-1}{d-2} \binom{s-d_1}{d-1} + \binom{d_1-1}{d-1} \binom{s-d_1-1}{d-2} \right].$$

Proof 1. Let us start with the lower bound. First, we focus on the polynomial $x^i \tilde{h}_0$.

Due to Lemma 1 it can be constructed in

$$r \sum_{v=1}^d \sum_{\substack{j_1, \dots, j_v \mid j_i > 0 \ \forall i \\ j_1 + \dots + j_v = r-d}} \binom{d-1}{v-1} \frac{1}{v}$$

ways, where v represents the number of runs of ones and the tuples $(j_1, \dots, j_v)$ represent the length of the runs of zero.

Let us now focus on the polynomial $x^i \tilde{h}_1$. Notice that it must be chosen such that $\deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r$, and consequently, the degree of $\tilde{h}_1$ must be at most $r - \deg(\tilde{h}_0) - 1$. This is equivalent to requiring that all the ones are within $r - \deg(\tilde{h}_0)$ consecutive positions of the circular register. The degree of the polynomial $\tilde{h}_0$ is determined by the tuple $(j_1, \dots, j_v)$. As a consequence, the degree of $\tilde{h}_1$ has to be at most j_k as k varies, depending on the position on the register where the first coefficient of $\tilde{h}_1$ is placed (or equivalently depending on the shift x^i). Let $m = \max\{j_i \mid i \in \{1, \dots, v\}\}$.

For every polynomial $x^i \tilde{h}_0$ a lower bound to the number of polynomials $x^i \tilde{h}_1$ such that $(x^i \tilde{h}_0, x^i \tilde{h}_1) \in \mathcal{W}_1$ is

$$\binom{m+1}{d}.$$

The latter represents the number of polynomials of weight d and degree at most m . Consequently, we can state that

$$|\mathcal{W}_1| \geq r \sum_{v=1}^d \sum_{\substack{j_1, \dots, j_v \mid j_i > 0 \ \forall i \\ j_1 + \dots + j_v = r-d}} \binom{d-1}{v-1} \frac{1}{v} \binom{m+1}{d}.$$

As in Section 5.1.1, we can reformulate this equation into the more easily computable form

$$|\mathcal{W}_1| \geq r \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v} \binom{m+1}{d},$$

which is the claimed lower bound.

2. Let us turn our attention to the upper bound. $\mathcal{W}_1$ can be written as

$$\mathcal{W}_1 = \bigcup_{s=2d-2}^{r-1} \mathcal{W}_1^s,$$

where

$$\mathcal{W}_1^s := \{(x^i \tilde{h}_0, x^i \tilde{h}_1) \in \mathcal{R}^2 \mid \deg(\tilde{h}_0) + \deg(\tilde{h}_1) = s, \\ |\tilde{h}_0| = |\tilde{h}_1| = d, \ \tilde{h}_0(0) \vee \tilde{h}_1(0) = 1, \ i \in \mathbb{Z}_r\}.$$

Note that, since the sets $\mathcal{W}_1^s$ are not disjoint,

$$|\mathcal{W}_1| \leq \sum_{s=2d-2}^{r-1} |\mathcal{W}_1^s|.$$

In turn, each set $\mathcal{W}_1^s$ can be seen as the union of the sets $\mathcal{W}_1^{s,i}$, with $i \in \mathbb{Z}_r$, defined as

$$\mathcal{W}_1^{s,i} := \{(x^i \tilde{h}_0, x^i \tilde{h}_1) \in \mathcal{R}^2 \mid \deg(\tilde{h}_0) + \deg(\tilde{h}_1) = s, \\ |\tilde{h}_0| = |\tilde{h}_1| = d, \tilde{h}_0(0) \vee \tilde{h}_1(0) = 1\}.$$

Note that $\mathcal{W}_1^{s,i}$ represents the subset of $\mathcal{W}_1^s$ where the shift i is fixed. Consequently,

$$\mathcal{W}_1 = \bigcup_{s=2d-2}^{r-1} \mathcal{W}_1^s = \bigcup_{s=2d-2}^{r-1} \bigcup_{i=0}^{r-1} \mathcal{W}_1^{s,i}.$$

Since $\forall i_1, i_2 \in \mathbb{Z}_r$ it holds that $|\mathcal{W}_1^{s,i_1}| = |\mathcal{W}_1^{s,i_2}|$, then

$$|\mathcal{W}_1| \leq \sum_{s=2d-2}^{r-1} |\mathcal{W}_1^s| \leq \sum_{s=2d-2}^{r-1} \sum_{i=0}^{r-1} |\mathcal{W}_1^{s,i}| \leq r \sum_{s=2d-2}^{r-1} |\mathcal{W}_1^{s,0}|.$$

To determine the cardinality of $\mathcal{W}_1^{s,0}$, we begin by noting that the degree of the first polynomial can range from $d-1$ to $s-(d-1)$. Thus, we sum over a variable d_1 spanning over these values.

Considering that at least one of the two polynomials must have a constant term set to 1, we identify two cases:

- (a) The first polynomial has a constant term equal to 1;
- (b) The first polynomial has a constant term equal to 0, leading to the second polynomial having a constant term set to 1.

Consequently, let d_1 represent the degree of the first polynomial,

$$|\mathcal{W}_1^{s,0}| = \sum_{d_1=d-1}^{s-(d-1)} \left[\binom{d_1-1}{d-2} \binom{s-d_1}{d-1} + \binom{d_1-1}{d-1} \binom{s-d_1-1}{d-2} \right].$$

In conclusion, we can state that

$$|\mathcal{W}_1| \leq r \sum_{s=2d-2}^{r-1} \sum_{d_1=d-1}^{s-(d-1)} \left[\binom{d_1-1}{d-2} \binom{s-d_1}{d-1} + \binom{d_1-1}{d-1} \binom{s-d_1-1}{d-2} \right].$$

□

At this stage, we have the exact cardinality of $\mathcal{W}_2$ and an estimate for that of $\mathcal{W}_1$. We want to obtain similar results for K_1 and K_2 , hence, we need to also consider the coprimality requirement.

5.2 Cardinality of the sets K_1 and K_2

In this section we describe the process used to estimate the cardinality of K_1 and K_2 , starting from $\mathcal{W}_1$ and $\mathcal{W}_2$. This estimation was carried out through a statistical analysis aimed at determining the proportion of elements in $\mathcal{W}_1$ and $\mathcal{W}_2$ that also belong to K_1 and K_2 , respectively. Also in this section we will start with case 2, followed by case 1.

5.2.1 Case 2:

We want to know the probability that a pair of polynomials in $\mathcal{W}_2$ is also inside K_2 , that is, the probability that it can be written as $(x^i \tilde{h}_0, x^j \tilde{h}_1)$ with $\tilde{h}_0$ and $\tilde{h}_1$ coprime in $\mathbb{F}_2[x]$.

Note that, since the pairs of polynomials in $\mathcal{W}_2$ live in the ring $\mathcal{R}$, there is not a unique representation of them. However, we can state that a pair is weak if it admits a representation of the just described type.

The analysis is performed using statistical methods. In this phase, the main challenge is to determine how to randomly select a pair of polynomials from $\mathcal{W}_2$. Note that, a straightforward approach would be to randomly generate a pair of polynomials in $\mathcal{P}_d$ and retain it only if it belongs to $\mathcal{W}_2$. However, this method could not possibly work, because the probability of a randomly generated pair falling within $\mathcal{W}_2$ becomes too small for large values of r , making it unfeasible to generate enough pairs to conduct a meaningful analysis.

On the other hand, generating all elements inside $\mathcal{W}_2$ and then selecting one at random is infeasible, as the set is too large and the memory requirements to carry out the random generation in this way would be prohibitive.

In light of these thoughts, we will proceed by designing a method to construct the elements of $\mathcal{W}_2$ as if they were randomly sampled.

Let us define the set

$$\tilde{\mathcal{W}}_2 := \{(h_0, h_1) \in \mathcal{R}^2 \mid |h_0| = |h_1| = d, \deg(h_0) + \deg(h_1) < r, h_0(0) = h_1(0) = 1, \\ \nexists (h'_0, h'_1) \in \tilde{\mathcal{W}}_2 \text{ and } i, j \in \mathbb{Z}_r \text{ s.t. } (x^i h_0, x^j h_1) = (h'_0, h'_1)\}. \quad (6)$$

Formally speaking, an equivalence relation is defined on $\mathcal{W}_2$, such that two pairs of polynomials are equivalent if one can be obtained from the other by applying shifts to both the first and second polynomial. The set $\tilde{\mathcal{W}}_2$ is simply a set of representatives for these equivalence classes.

Hence, by conducting a statistical test to approximate the probability that a pair of polynomials in $\tilde{\mathcal{W}}_2$ is coprime in $\mathbb{F}_2[x]$, we can establish a lower bound on the

probability that a pair of polynomials in $\mathcal{W}_2$ is a weak pair.

First, we need to discuss how to select elements from $\tilde{\mathcal{W}}_2$ according to the uniform distribution. This requires studying certain properties of polynomials uniformly sampled from $\tilde{\mathcal{W}}_2$. We remark that, indeed, not all features within $\tilde{\mathcal{W}}_2$ are distributed uniformly: for example, the length of the longest sequence of zeros in one polynomial of a uniformly picked pair may appear more frequently within certain ranges than in others. Our idea is to leverage these properties in order to define an efficient procedure that outputs a polynomial uniformly selected from $\tilde{\mathcal{W}}_2$.

Procedure 1

1. **Picking the longest zeros run in h_0 :** we select a specific value for m , called $\bar{m}$, based on $\mathcal{D}_1$;
2. **Selecting the number of runs of length $\bar{m}$:** as a second step, we determine a certain value for the number n of runs of length $\bar{m}$ within h_0 . Now, by relying on the probability distribution $\mathcal{D}_2^{\bar{m}}$, we can pick a certain $\bar{n}$;
3. **Selecting the remaining zeros runs:** now, we need to determine the lengths of the remaining zeros runs in h_0 . We randomly pick a subset of size $d - \bar{n} - 1$ from the set $\{2, \dots, r - \bar{n}(\bar{m} + 1)\}$. Think of this subset as markers separating the zeros runs. Each marker must be at most $\bar{m} - 1$ positions away from the previous and next one. The first separator needs to be at most $\bar{m} - 1$ positions away from the position 1 and the last separator needs to be at most $\bar{m} - 1$ positions away from the position $r - \bar{n}(\bar{m} + 1) + 1$. If any of these checks fail, we simply discard the randomly generated subset and try again. In this way we create $d - \bar{n}$ zeros runs with length between 0 and $\bar{m} - 1$ (note that a run of length 0 is not a real run);
4. **Creating the polynomial h_0 :** we start by randomly distributing $\bar{n} - 1$ longest zeros runs among the $d - \bar{n}$ zeros runs identified in the previous step (one longest run is kept aside, to maintain it at the end). To do it we generate a sequence L of length $d - 1$. Then, we randomly select $\bar{n} - 1$ positions from $\{1, \dots, d - 1\}$ and set these positions in the sequence to the value $\bar{m}$. The remaining entries are filled with the lengths of the other runs. Finally, we create h_0 : a polynomial of degree $r - \bar{m} - 1$ that has constant term equal to 1, such that its zeros runs precisely match the entries of the sequence L ;
5. **Picking the longest run in h_1 :** we pick up a value $\bar{k} \in \{r - (\bar{m} + 1), \dots, r - d\}$ for k , representing the longest zeros run in h_1 , by using $\mathcal{D}_3^{\bar{m}}$;
6. **Counting the runs of length $\bar{k}$:** we select the number s of runs of length $\bar{k}$ within h_1 , using $\mathcal{D}_2^{\bar{k}}$. We will call this value $\bar{s}$;
7. **Creating the polynomial h_1 :** finally, we create the polynomial h_1 using the same approach utilized for h_0 , but with $\bar{k}$ and $\bar{s}$ instead of $\bar{m}$ and $\bar{n}$, respectively.

$\mathcal{D}_1$ represents the probability distribution of the length of the longest zero run in h_0 , while $\mathcal{D}_2^m$ represents the probability distribution to have exactly n longest runs of zeros of length m , once m is fixed. $\mathcal{D}_3^m$ represents the probability distribution of the length of the longest zero run in h_1 , once the longest zero run of h_0 is set to m .

In preparation for the proof of correctness, we present some preliminary lemmas.

Lemma 4 *Let $\tilde{\mathcal{W}}_2$ be the set described by Equation (6), then*

$$|\tilde{\mathcal{W}}_2| = \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v} \cdot \left(\sum_{k=1}^d \left(\binom{r-d-1}{k-1} - R(r-d-k, k, r-m-3) \right) \binom{d-1}{k-1} \frac{1}{k} \right).$$

Proof Directly derived from Theorem 3. $\square$

Lemma 5 *For a fixed value of m and n , the number of pairs of polynomials (h_0, h_1) in $\tilde{\mathcal{W}}_2$ such that h_0 has exactly n longest zeros runs of value m is*

$$R(r-d-mn, d-n, m-1) \binom{d}{n} \frac{1}{d}.$$

Proof Once that n is fixed, we are left with at most $d-n$ zeros runs. For simplicity, we say that we have exactly $d-n$ runs left, each taking values between 0 and $m-1$, and collectively summing up to $r-d-mn$. Note that, in this case, a run of length 0 is not a real run. The manner in which these remaining runs can be selected mirrors the process of distributing $r-d-mn$ balls into $d-n$ bins, each with a capacity of $m-1$. At this point, the result must be multiplied by the number of ways in which the n runs of maximum length can be positioned among the other runs. This is given by the binomial coefficient $\binom{d}{n}$. The final detail to address is to avoid circular repetitions. Consequently, the result has to be divided by d . $\square$

At this point we can state the following theorem

Theorem 5 *The output of Procedure 1 is a pair of polynomials uniformly drawn from $\tilde{\mathcal{W}}_2$.*

Proof The two polynomials h_0 and h_1 will be represented by two registers of length r with the first position set to one and the longest sequence of zeros placed at the end.

Let us consider h_0 . The probability distribution $\mathcal{D}_1$ of its longest sequence of zeros having a specific value m within the range $[\lceil \frac{r-d}{d} \rceil, \dots, r-d]$ is not uniform. To characterize $\mathcal{D}_1$, we can tally, for each value of m , the number of pairs in $\tilde{\mathcal{W}}_2$ where h_0 has a longest zeros run of length m , and, then, normalize it. To do this, we can rely on Lemma 4 that describes the cardinality of the set, where the variable m represents the longest sequence of zeros in the first polynomial.

At this point, for every possible value of m , we also need to know the probability for h_0 to have exactly n runs of zeros of length m . This will help us in assigning equal probability to each equivalence class of polynomials. These probability distributions, that depend on

the fixed value of m are called $\mathcal{D}_2^m$. Let us fix m . To find $\mathcal{D}_2^m$ we have to count, for every value of n in $\{1, \dots, d\}$, the pair of polynomials (h_0, h_1) in $\mathcal{W}_2$ such that h_0 has exactly n longest zeros runs of value m . For a fixed m , once the number of pairs in $\mathcal{W}_2$ with n runs of maximum length m in the first polynomial has been determined, following Lemma 5, we simply normalize these values. This yields the desired probability distributions $\mathcal{D}_2^m$.

Now, let us turn our attention to the second polynomial of the pair: h_1 . Once h_0 is fixed, the options for h_1 become more restricted. To generate the polynomial h_1 such that the pair (h_0, h_1) appears as if it were randomly drawn from $\mathcal{W}_2$, additional probability distributions must be determined.

First, for each possible longest zeros run length (denoted by m) in h_0 , we need to characterize the probability distribution for the longest zeros run in h_1 . These probability distributions will be denoted with $\mathcal{D}_3^m$. The value of the longest zeros run in h_1 , that we call k , can range from $r - (m + 1)$ all the way up to $r - d$. To determine the number of polynomials h_1 with a specific k value, we can follow the same logic used for calculating the cardinality of $\mathcal{W}_2$ in Theorem 3. Essentially, for each possible k between $r - (m + 1)$ and $r - d$, the number of polynomials h_1 with a longest zeros run of length k is given by

$$\sum_{v=1}^d (R(r - d - v, v, k - 1) - R(r - d - v, v, k - 2)) \binom{d-1}{v-1} \frac{1}{v}.$$

Once we have counted, for a fixed m , the number of polynomials h_1 for each possible k , we simply have to normalize these values in order to obtain $\mathcal{D}_3^m$. For a fixed k value, we might still expect to determine a separate probability distribution for the number of runs of maximum length k within h_1 . However, this distribution is actually identical to the one previously calculated for the number of runs of maximum length m in h_0 . □

5.2.2 Case 1

At this point we want to know the probability that a pair of polynomial inside $\mathcal{W}_1$ is also inside K_1 .

In order to conduct this analysis we have to randomly sample some pairs in $\mathcal{W}_1$ and check if there exist a representation of them of the type $(x^i \tilde{h}_0, x^i \tilde{h}_1)$ such that $\deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r$ and $\gcd(\tilde{h}_0, \tilde{h}_1) = 1$ in $\mathbb{F}_2[x]$.

However, in this scenario, generating random polynomial pairs from $\mathcal{W}_1$ is even more challenging than the previous case, because we lack a precise formula for the cardinality of $\mathcal{W}_1$.

We have therefore decided to start from the upper and lower bounds established in Section 5.1.2. Since these bounds represent, respectively, the cardinality of a subset and of a multiset¹ of elements within $\mathcal{W}_1$, we conduct our statistical analysis on these sets instead of on $\mathcal{W}_1$ itself.

Let us start from the lower bound and consider the following subset of $\mathcal{W}_1$

$$\mathcal{W}_1^{\text{low}} := \{(x^i h_0, x^i h_1) \in \mathcal{R}^2 \mid |h_0| = |h_1| = d, r - \deg(h_0) > \text{each zeros run of } h_0, h_0(0) = 1, \deg(h_0) + \deg(h_1) < r, i \in \mathbb{Z}_r\}. \quad (7)$$

¹a multiset is a generalization of a set, where repetition of elements is allowed

Lemma 6 Let $\mathcal{W}_1^{low}$ be the set defined in Equation (7), then

$$|\mathcal{W}_1^{low}| = r \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v} \binom{m+1}{d}.$$

Proof Directly derived from Theorem 4 point 1. $\square$

Corollary 1 Let

$$\tilde{\mathcal{W}}_1^{low} := \{(h_0, h_1) \in \mathcal{R}^2 \mid |h_0| = |h_1| = d, r - \deg(h_0) > \text{each zeros run of } h_0, h_0(0) = 1, \\ \deg(h_0) + \deg(h_1) < r, \nexists (h'_0, h'_1) \in \tilde{\mathcal{W}}_1^{low} \text{ and } i \in \mathbb{Z}_r \text{ s.t. } (x^i h_0, x^i h_1) = (h'_0, h'_1)\}.$$

Then

$$|\tilde{\mathcal{W}}_1^{low}| = \sum_{v=1}^d \sum_{m=\lceil \frac{r-d}{v} \rceil}^{r-d-(v-1)} (R(r-d-v, v, m-1) - R(r-d-v, v, m-2)) \binom{d-1}{v-1} \frac{1}{v} \binom{m+1}{d}.$$

Proof Focusing on the factor r in the description of the cardinality of $\mathcal{W}_1^{low}$ in Lemma 6, we can note that the set $\mathcal{W}_1^{low}$ consists of the disjoint union of the set $\tilde{\mathcal{W}}_1^{low}$ with the set of all pairs of polynomials in $\tilde{\mathcal{W}}_1^{low}$ subjected to an equal shift. More formally, we can define an equivalence relation on $\mathcal{W}_1^{low}$, such that two pairs of polynomials are equivalent if one can be obtained from the other by applying an equal shift to both polynomials. The set $\tilde{\mathcal{W}}_1^{low}$ is a set of representatives for these equivalence classes. From this we get the statement of the Corollary. $\square$

We want to conduct an analysis on the probability that a pair of polynomials in $\mathcal{W}_1^{low}$ can be represented as $(x^i \tilde{h}_0, x^i \tilde{h}_1)$, with $\deg(\tilde{h}_0) + \deg(\tilde{h}_1) < r$ and $\gcd(\tilde{h}_0, \tilde{h}_1) = 1$ in $\mathbb{F}_2[x]$.

We will conduct our analysis on the set $\tilde{\mathcal{W}}_1^{low}$ defined in Corollary 1. However, since the polynomial pairs we are working with reside in the quotient ring $\mathcal{R}$, they may have multiple representations of the form $(x^i \tilde{h}_0, x^i \tilde{h}_1)$. Therefore, when analyzing $\tilde{\mathcal{W}}_1^{low}$, we account for only one representation of each pair. This approach allows us to derive a lower bound for the probability that a pair in $\mathcal{W}_1^{low}$ is weak. Let us define a procedure that mimics a random draw from $\tilde{\mathcal{W}}_1^{low}$.

Procedure 2

1. **Selecting the length of the longest zeros run in $\mathbf{h}_0$** , that we will call $\tilde{m}$;

2. **Determining the number of runs of length $\bar{m}$ inside h_0** , that we will denote with $\bar{n}$;
3. **Selecting the remaining zeros runs**;
4. **Creating the polynomial h_0** ;

These first four steps are performed exactly as in Section 5.2.1, utilizing $\mathcal{D}_4$, instead of $\mathcal{D}_1$, and $\mathcal{D}_2^{\bar{m}}$. $\mathcal{D}_4$ represents the probability distribution of the longest zeros run in the polynomial h_0 and $\mathcal{D}_2^{\bar{m}}$ represents the probability distribution for h_0 to have exactly n longest zeros runs of length m , once m is fixed.

5. **Creating the polynomial h_1** : to create this second polynomial we just need to generate an r -long sequence of zeros and randomly select d positions among the first $\bar{m} + 1$, whose values are set to 1. This resulting sequence represents the coefficients of h_1 .

Theorem 6 *The output of Procedure 2 is a pair of polynomials uniformly drawn from $\tilde{\mathcal{W}}_1^{\text{low}}$.*

Proof Firstly, we focus on the polynomial h_0 , which can be represented by an r -long register with the first position set to 1 and the longest zeros run placed at the end.

The probability distributions that we will use to generate a pair (h_0, h_1) , as if it were randomly drawn from $\tilde{\mathcal{W}}_1^{\text{low}}$, can be computed in the same way as in Section 5.2.1.

More specifically, the cardinality formula for the set $\tilde{\mathcal{W}}_1^{\text{low}}$, stated in Corollary 1, enables efficient enumeration of elements where the first polynomial has a longest zeros run of length m , with m ranging from $\lceil \frac{r-d}{d} \rceil$ to $r-d$. This enumeration, followed by normalization, computes the probability distribution of the longest zeros run in the polynomial h_0 , that we will call $\mathcal{D}_4$.

Next, for each possible value of m , we also need to determine the probability distribution for h_0 having exactly n zeros runs of length m . These distributions, which differ depending on the fixed value of m , are exactly $\mathcal{D}_2^m$, presented in Procedure 1.

At this point we can focus on h_1 , which must satisfy the condition of having degree at most m . $\square$

Regarding the upper bound of Theorem 4, the situation is different. Unlike the cases just discussed, the bound we found is not only an upper bound for $|\mathcal{W}_1|$, but also for $|K_1|$, since $|K_1| \leq |\mathcal{W}_1|$.

Consequently, we can define a set that contains K_1 and has cardinality matching the upper bound, and thus conduct a statistical analysis to determine a lower bound on the probability that the elements of this superset are coprime. We remark that this approach does not hold the same relevance as the analyses conducted so far. However, it can still provide insight into the impact of the coprimality requirement on the count of weak pairs. For this reason, we have decided to perform this statistical analysis as well. Let us define the following set

$$\tilde{\mathcal{W}}_1^{\text{up}} := \{(h_0, h_1) \in \mathcal{R}^2 \mid |h_0| = |h_1| = d, h_0(0) \vee h_1(0) = 1, \deg(h_0) + \deg(h_1) < r\}. \quad (8)$$

Let $\mathcal{W}_1^{\text{up}}$ denote the multiset formed by taking the just defined set and by adding all elements in $\tilde{\mathcal{W}}_1^{\text{up}}$ to which an equal shift to both polynomials is applied. Note that $\mathcal{W}_1^{\text{up}}$ is a multiset, and not a simple set, since it contains some pairs more than one time. What is essential, however, is that $\mathcal{W}_1^{\text{up}}$ contains all elements of $\mathcal{W}_1$.

The probability that a pair in $\mathcal{W}_1^{\text{up}}$ is weak is not equal to the probability that a pair in $\tilde{\mathcal{W}}_1^{\text{up}}$ is coprime. We can however say that the second probability is a lower bound for the first one.

Lemma 7 *Let $\tilde{\mathcal{W}}_1^{\text{up}}$ be the set defined in Equation (8). Then,*

$$|\tilde{\mathcal{W}}_1^{\text{up}}| = \sum_{s=2d-2}^{r-1} \sum_{d_1=d-1}^{s-(d-1)} \left[\binom{d_1-1}{d-2} \binom{s-d_1}{d-1} + \binom{d_1-1}{d-1} \binom{s-d_1-1}{d-2} \right],$$

where the variable s represents the sum of the degrees of the two polynomials, and the variable d_1 represents the degree of the first polynomial.

Proof Directly derived from Theorem 4 point 2. □

We will now focus on the set $\tilde{\mathcal{W}}_1^{\text{up}}$. We want to conduct a statistical analysis to estimate the probability that a pair of polynomials inside this set is coprime in $\mathbb{F}_2[x]$. To achieve this, we need a method for randomly selecting elements from the set. Also in this case, in order to efficiently construct elements of $\tilde{\mathcal{W}}_1^{\text{up}}$ as if they were randomly picked, we need to focus on some properties of the two polynomials and find their probability distributions.

The procedure we will use for constructing elements of $\tilde{\mathcal{W}}_1^{\text{up}}$ as if they were randomly selected is the following.

Procedure 3

1. **Selecting the degree d_1 of the first polynomial:** we pick a value $\bar{d}_1$, for the degree of h_0 , following the probability distribution $\mathcal{D}_5$;
2. **Generating the polynomial h_0 :** we randomly generate a polynomial of degree $\bar{d}_1$ and of weight d . To do this we build a sequence of $\bar{d}_1 + 1$ zeros, that represents the coefficients of the polynomials. Then, we turn the coefficient in position $\bar{d}_1 + 1$ to 1, in this way the resulting polynomial will be of degree $\bar{d}_1$. We then select randomly other $d - 1$ positions among the remaining ones and we set these coefficients to 1;
3. **Generating the polynomial h_1 :** we randomly generate a polynomial of degree at most $r - 1 - \bar{d}_1$ and weight d , following a similar method used in the previous point. We build a sequence of $r - 1 - \bar{d}_1$ zeros that represents the coefficients and we randomly select d positions that are turned to 1;

4. **Checking if at least one among h_0 and h_1 has constant term equal to 1:**
 we check if the condition $h_0(0) \vee h_1(0) = 1$ is satisfied, otherwise we discard h_0 and h_1 and restart from the second step.

$\mathcal{D}_5$ represents the probability distribution of the degree of h_0 .

Theorem 7 *The output of Procedure 3 is a pair of polynomials uniformly drawn from $\tilde{\mathcal{W}}_1^{up}$.*

Proof Let us start by focusing on the first polynomial of the pair. Analyzing the formula that describes $|\tilde{\mathcal{W}}_1^{up}|$, given in Lemma 7, the probability distribution $\mathcal{D}_5$ of the degree of h_0 can be easily found.

At this point the second polynomial must be of degree at most $r - 1 - \deg(h_0)$ and at least one among h_0 and h_1 must have constant term equal to 1. $\square$

5.2.3 Experimental setup

Now that we have constructed a method to efficiently generate random polynomial pairs in $\tilde{\mathcal{W}}_2$, $\tilde{\mathcal{W}}_1^{\text{low}}$ and $\tilde{\mathcal{W}}_1^{up}$, we can proceed with the statistical analysis. To do this we need to fix the parameters r and d .

Since Problem 1 is closely related to the security of certain post-quantum code-based schemes, solving it may, in some cases, lead to breaking the corresponding cryptosystems. To highlight this connection and carry out a statistical analysis with real-world parameters, in the next section we apply our approach to a known scheme.

6 Key-recovery attack for BIKE

BIKE [4] is a Key-Encapsulation Mechanism (KEM) based on the Niederreiter framework [15] that uses QC-MDPC codes. We briefly present here the parameters of BIKE, the key generation algorithm, and, thus, the link between the public and private keys. For a complete description of BIKE we refer to Appendix A.

BIKE uses QC-MDPC codes of length $n = 2r$, dimension r , and designed minimum distance w . In particular, r is a positive prime integer with the property that 2 is primitive modulo r , that determines the structure of the polynomial ring $\mathcal{R}$. $w \approx \sqrt{n}$ is an even positive integer used to define the row weight of the QC-MD parity-check matrix of the code, and it has to be such that $d := w/2$ is odd. Other than r and w , the parameters in BIKE include t and l , which however do not have an impact on the key generation and thus on our analysis. The parameters suggested in BIKE to achieve the three NIST security categories are shown in Table 1.

For the scope of this work, we will focus only on the main points of the key generation algorithm.

The private key consists of a pair of polynomials (h_0, h_1) , randomly selected from the quotient ring $\mathcal{R}$, each of weight exactly d , representing a QC-MDPC matrix, namely, the private keys' space $\mathcal{H}_w$ of BIKE is the set $\{(h_0, h_1) \in \mathcal{R}^2 \mid |h_0| = |h_1| = d\}$. The public key $h := h_1 h_0^{-1} \in \mathcal{R}$ represents an equivalent QC parity-check matrix.

Security level	Parameters			
	r	w	t	l
I	12 323	142	134	256
III	24 659	206	199	256
V	40 973	274	264	256

Table 1: Suggested BIKE parameters

Note that recovering the two polynomials that constitute the private key, starting from the public key, means solving Problem 1.

Consequently, the presented approaches constitute some key-recovery attacks for BIKE, and the sets K_1 and K_2 represent the weak keys retrievable with these two attacks. From now on, we will refer to as Attack 1 the strategy where we attempt to find the private key by running the MEEA on $(x^r - 1, h, d)$, while Attack 2 consists of running the MEEA on each shift of h . In order to estimate the cost of these two attacks we need to know the probability of selecting a weak key. This probability is the rate between the cardinality of the sets K_1 (or K_2), and the cardinality of the weak key space $\mathcal{H}_w$ (i.e. $\mathcal{P}_d$). In order to estimate $|K_1|$ and $|K_2|$, we proceed first by computing $|\mathcal{W}_1|$ and $|\mathcal{W}_2|$. Then, using MAGMA to generate the polynomials according to Section 5.2.1 and Section 5.2.2, we perform the statistical analysis.

6.1 Experimental results

We estimate the cardinality of $|K_1|$ and $|K_2|$ with the parameters suggested for the I and III security level of BIKE.

To reliably estimate the probability that a pair inside $\tilde{\mathcal{W}}_2$, $\tilde{\mathcal{W}}_1^{\text{low}}$ or $\tilde{\mathcal{W}}_1^{\text{up}}$ is coprime we will calculate the corresponding 99% confidence intervals (CI). For a more complete description of the construction of CI see Appendix B.

I-security level parameters

In this section we will use the first-security level parameters presented in Table 1. For all the analysis conducted with these parameters we collect a sample of size $n = 10^5$. This large sample size contributes to a narrower CI, leading to a more precise estimation of the coprimality probability. We set a confidence level of 99% ($1 - \alpha = 0.99$).

Let us start with the randomly chosen polynomial pairs of $\tilde{\mathcal{W}}_2$, obtained by using Procedure 1. Out of these 100 000 pairs, 89 083 are coprime. Consequently

$$\bar{X}_n = \frac{89\,083}{100\,000} = 0.89083,$$

$$S_n^2 = \frac{89\,083(1 - 0.89083)^2 + 10\,917(0 - 0.89083)^2}{99\,999} = 0.09725,$$

$$\begin{aligned}\text{CI}_p(0.99) &= \left[0.89083 - 2.576\sqrt{\frac{0.09725}{100\,000}}; 0.89083 + 2.576\sqrt{\frac{0.09725}{100\,000}} \right] \\ &= [0.88828; 0.89337].\end{aligned}$$

According to this analysis we can say that, with a 99% level of confidence, the pairs (h_0, h_1) in $\tilde{\mathcal{W}}_2$ such that h_0 and h_1 are coprime in $\mathbb{F}_2[x]$ are between the 88.828% and the 89.337%.

It is worth remembering that this probability is a lower bound to the probability that a pair in $\mathcal{W}_2$ is a weak key.

Let us now consider $\tilde{\mathcal{W}}_1^{\text{low}}$ and $\tilde{\mathcal{W}}_1^{\text{up}}$. First of all let us focus on the lower bound. Out of 100 000 pairs randomly selected from $\tilde{\mathcal{W}}_1^{\text{low}}$ following Procedure 2, 88 857 are coprime in $\mathbb{F}_2[x]$, consequently, the sample mean and the sample variance are

$$\bar{X}_n = \frac{88\,857}{100\,000} = 0.88857,$$

$$S_n^2 = \frac{88\,857(1 - 0.88857)^2 + 11\,143(0 - 0.88857)^2}{99\,999} = 0.09901.$$

This yields the following confidence interval

$$\begin{aligned}\text{CI}_p(0.99) &= \left[0.88857 - 2.576\sqrt{\frac{0.09901}{100\,000}}; 0.88857 + 2.576\sqrt{\frac{0.09901}{100\,000}} \right] \\ &= [0.88600; 0.89113].\end{aligned}$$

Hence, we can say with a 99% level of confidence that, between the 88.600% and 89.113% of pairs (h_0, h_1) in the set $\tilde{\mathcal{W}}_1^{\text{low}}$ are coprime in $\mathbb{F}_2[x]$.

As already mentioned, this value is a lower bound for the probability that an element in $\mathcal{W}_1^{\text{low}}$ is a weak key.

Let us now focus on the upper bound case. Out of 100 000 pairs inside $\tilde{\mathcal{W}}_1^{\text{up}}$ sampled following Procedure 3, 88 807 are coprime. This translates into the following sample mean and sample variance,

$$\bar{X}_n = \frac{88\,807}{100\,000} = 0.88807,$$

$$S_n^2 = \frac{88\,807(1 - 0.88807)^2 + 11\,193(0 - 0.88807)^2}{99\,999} = 0.09940,$$

and into the following confidence interval

$$\begin{aligned} \text{CI}_p(0.99) &= \left[0.88807 - 2.576\sqrt{\frac{0.09940}{100\ 000}}; 0.88807 + 2.576\sqrt{\frac{0.09940}{100\ 000}} \right] \\ &= [0.88550; 0.89064]. \end{aligned}$$

Hence, with a high probability, within the set $\tilde{\mathcal{W}}_1^{\text{up}}$, between 88.550% and 89.064% of pairs are coprime in $\mathbb{F}_2[x]$. Consequently, it is likely than more that 88% of the pairs in $\mathcal{W}_1^{\text{up}}$ are weak keys (referring to the proportion of pairs within the entire collection, including duplicates). Given such a high percentage, attempting to establish an upper bound for the probability that an element in $\mathcal{W}_1^{\text{up}}$ can be represented as $(x^i h_0, x^i h_1)$ such that $\deg(h_0) + \deg(h_1) < r$ and $\gcd(h_0, h_1) = 1$ in $\mathbb{F}_2[x]$ would probably not lead to a relevant improvement over the following bound:

$$|K_1| \leq |\mathcal{W}_1^{\text{up}}|.$$

In conclusion, with BIKE's first set of parameters $r = 12\ 323$ and $w = 142$, a conservative estimation is

$$\frac{1}{2} |\mathcal{W}_2| \leq |K_2| \qquad \frac{1}{2} |\mathcal{W}_1^{\text{low}}| \leq |K_1| \leq |\mathcal{W}_1^{\text{up}}|. \quad (9)$$

III-security level parameters

In this section, we set the parameters according to those suggested for the third-security level, as presented in Table 1. For the analysis conducted with these parameters, the size of the selected sample is $n = 10^4$, and the confidence level is set at 99%.

First, let us start with the set $\tilde{\mathcal{W}}_2$. Out of the 10 000 pairs randomly selected by using Procedure 1, 8 904 are coprime in $\mathbb{F}_2[x]$. Hence, the sample mean and the sample variance estimators are respectively given by

$$\bar{X}_n = \frac{8\ 904}{10\ 000} = 0.8904,$$

$$S_n^2 = \frac{8\ 904(1 - 0.8904)^2 + 1\ 096(0 - 0.8904)^2}{9\ 999} = 0.0976,$$

yielding the following confidence interval

$$\begin{aligned} \text{CI}_p(0.99) &= \left[0.8904 - 2.576\sqrt{\frac{0.0976}{10\ 000}}; 0.8904 + 2.576\sqrt{\frac{0.0976}{10\ 000}} \right] \\ &= [0.8823; 0.8985]. \end{aligned}$$

Let us now consider $\tilde{\mathcal{W}}_1^{\text{low}}$ and $\tilde{\mathcal{W}}_1^{\text{up}}$. Firstly, let us focus on the lower bound case. Out of 10 000 pairs, sampled from $\tilde{\mathcal{W}}_1^{\text{low}}$ by following Procedure 2, 8 840 are coprime, consequently

$$\begin{aligned}\bar{X}_n &= \frac{8\,840}{10\,000} = 0.8840, \\ S_n^2 &= \frac{8\,840(1 - 0.8840)^2 + 1\,160(0 - 0.8840)^2}{9\,999} = 0.1026, \\ \text{CI}_p(0.99) &= \left[0.8840 - 2.576\sqrt{\frac{0.1026}{10\,000}}; 0.8840 + 2.576\sqrt{\frac{0.1026}{10\,000}} \right] \\ &= [0.8757; 0.8923].\end{aligned}$$

For the upper bound case we get that out of 10 000 pairs, randomly selected from $\tilde{\mathcal{W}}_1^{\text{up}}$ by using Procedure 3, 8 867 are coprime. Hence,

$$\begin{aligned}\bar{X}_n &= \frac{8\,867}{10\,000} = 0.8867, \\ S_n^2 &= \frac{8\,867(1 - 0.8867)^2 + 1\,133(0 - 0.8867)^2}{9\,999} = 0.1004. \\ \text{CI}_p(0.99) &= \left[0.8867 - 2.576\sqrt{\frac{0.1004}{10\,000}}; 0.8867 + 2.576\sqrt{\frac{0.1004}{10\,000}} \right] \\ &= [0.8785; 0.8949].\end{aligned}$$

Consequently, also in this case attempting to establish an upper bound for the probability that an element in $\mathcal{W}_1^{\text{up}}$ can be represented as $(x^i h_0, x^i h_1)$ such that $\deg(h_0) + \deg(h_1) < r$ and $\gcd(h_0, h_1) = 1$ in $\mathbb{F}_2[x]$ would probably not lead to a relevant improvement.

In conclusion with BIKE's third level parameters $r = 24\,659$ and $w = 206$, a conservative estimation is

$$\frac{1}{2}|\mathcal{W}_2| \leq |K_2| \qquad \frac{1}{2}|\mathcal{W}_1^{\text{low}}| \leq |K_1| \leq |\mathcal{W}_1^{\text{up}}|. \quad (10)$$

6.2 Attacks' cost

At this point, we can estimate the cost of the proposed attacks, bearing in mind that $\mathcal{H}_w$ represents the private key space.

- For the I-level of security we have that

$$|\mathcal{H}_w| = |\mathcal{P}_d| \approx 2^{1251.86};$$

- For the III-level of security we have that

$$|\mathcal{H}_w| = |\mathcal{P}_d| \approx 2^{1915.32}.$$

6.2.1 Attack 1

Let us now analyze the cost of the first attack that involves running Algorithm 2 on input $(x^r - 1, h, d)$.

The probability $\mathbb{P}_1$ of selecting a weak key is defined as the ratio between the number of private keys that can be broken through this attack and the total number of private keys, described in Equation (5). To be precise, this is a lower bound since we can not exclude the possibility that there exist keys breakable in this way outside K_1 , but we will consider it as an estimation.

I-security level parameters

Given the reasoning, experimental results, and discussion provided above, and considering the parameters set for the first security level, we obtain that

$$2^{1119.33} \approx \frac{1}{2} |\mathcal{W}_1^{\text{low}}| \leq |K_1| \leq |\mathcal{W}_1^{\text{up}}| \approx 2^{1121.33}.$$

Hence,

$$\frac{1}{2^{132.53}} \approx \frac{|\mathcal{W}_1^{\text{low}}|}{2 |\mathcal{H}_w|} \leq \mathbb{P}_1 \leq \frac{|\mathcal{W}_1^{\text{up}}|}{|\mathcal{H}_w|} \approx \frac{1}{2^{130.53}}.$$

Let us recall that, as stated in Section 4, the cost of Algorithm 2 in our scenario is

$$\mathbf{c}_{\text{Algorithm 2}} = \mathcal{O}(r^2) \approx 2^{27.18}.$$

In conclusion, the cost of the attack, defined as the ratio between the cost of Algorithm 2 and the probability that the attack is successful, i.e.

$$\mathcal{C}_{\text{Attack 1}} := \frac{\mathbf{c}_{\text{Algorithm 2}}}{\mathbb{P}_1},$$

can be estimated by the following bounds.

$$2^{158} \approx \frac{2^{27.18}}{\frac{1}{2^{130.53}}} \leq \mathcal{C}_{\text{Attack 1}} \leq \frac{2^{27.18}}{\frac{1}{2^{132.53}}} \approx 2^{160}.$$

III-security level parameters

Fixing the parameters to those advised for the third security level, we get that

$$2^{1719.57} \approx \frac{1}{2} |\mathcal{W}_1^{\text{low}}| \leq |K_1| \leq |\mathcal{W}_1^{\text{up}}| \approx 2^{1721.56}.$$

Consequently,

$$\frac{1}{2^{195.75}} \approx \frac{|\mathcal{W}_1^{\text{low}}|}{2^{|\mathcal{H}_w|}} \leq \mathbb{P}_1 \leq \frac{|\mathcal{W}_1^{\text{up}}|}{|\mathcal{H}_w|} \approx \frac{1}{2^{193.76}}.$$

Since the cost of Algorithm 2 is

$$\mathbf{c}_{\text{Algorithm 2}} = \mathcal{O}(r^2) \approx 2^{29.18},$$

we obtain

$$2^{223} \approx \frac{2^{29.18}}{\frac{1}{2^{193.76}}} \leq \mathcal{C}_{\text{Attack 1}} \leq \frac{2^{29.18}}{\frac{1}{2^{195.75}}} \approx 2^{225}.$$

6.2.2 Attack 2

Let us now evaluate the cost of the second attack, which requires executing Algorithm 3 on input $((x^r - 1, h), d)$.

It is worth noting that the cost of this second attack can not be lower than the first one presented in Section 6.2.1. This is because Algorithm 3 has a cost that is r times higher, while the number of weak keys that can be exploited by this attack does not increase proportionally by a factor of r . Nevertheless, for the sake of completeness, we will show in detail its cost.

The probability $\mathbb{P}_2$ of selecting a weak key for this case, defined as the ratio between the number of private key breakable with this attack and the cardinality of the private key space is described in Equation (5).

Note that, also in this case, this ratio is a lower bound estimation, but we will treat it as an approximation.

I-security level parameters

By fixing the parameters to those advised for the first level of security we get

$$2^{1126.47} \approx \frac{1}{2} |\mathcal{W}_2| \leq |K_2|.$$

Hence, the probability of selecting a weak key is bounded by the value

$$\frac{1}{2^{125.39}} \approx \frac{|\mathcal{W}_2|}{2^{|\mathcal{H}_w|}} \leq \mathbb{P}_2$$

and, as already seen, the cost of Algorithm 3 is estimated by

$$\mathbf{c}_{\text{Algorithm 3}} = \mathcal{O}(r^3) \approx 2^{40.77}.$$

In conclusion, as the cost of an attack is defined as the ratio between the resources needed to execute it and the probability of success, i.e.

$$\mathcal{C}_{\text{Attack 2}} := \frac{\mathbf{c}_{\text{Algorithm 3}}}{\mathbb{P}_2},$$

we obtain

$$\mathcal{C}_{\text{Attack 2}} \leq \frac{2^{40.77}}{\frac{1}{2^{125.39}}} \approx 2^{166}.$$

Despite this is an upper bound, we expect that the cost $\mathcal{C}_{\text{Attack}}$, as defined above, will not be too far from it.

III-security level parameters

By fixing the parameters to those advised for the third level of security, we get that

$$2^{1727.25} \approx \frac{1}{2} |\mathcal{W}_2| \leq |K_2|.$$

Hence,

$$\frac{1}{2^{188.07}} \approx \frac{|\mathcal{W}_2|}{2^{|\mathcal{H}_w|}} \leq \mathbb{P}_2.$$

The cost of Algorithm 3 is estimate by

$$\mathbf{c}_{\text{Algorithm 3}} = \mathcal{O}(r^3) \approx 2^{43.77}.$$

In conclusion an upper bound, that will not be too far from the real value, for the cost of this attack is

$$\mathcal{C}_{\text{Attack 2}} \leq \frac{2^{43.77}}{\frac{1}{2^{188.07}}} \approx 2^{232}.$$

6.3 Results

Table 2 summarizes the attack costs, in the specific setting where it is assumed that a weak key is used, and the probability of selecting a weak key.

Security level	Attack 1		Attack 2	
I	$2^{-132.53} \leq \mathbb{P} \leq 2^{-130.53}$	$\mathbf{c} \approx 2^{27.18}$	$2^{-125.39} \leq \mathbb{P}$	$\mathbf{c} \approx 2^{40.77}$
III	$2^{-195.75} \leq \mathbb{P} \leq 2^{-193.76}$	$\mathbf{c} \approx 2^{29.18}$	$2^{-188.07} \leq \mathbb{P}$	$\mathbf{c} \approx 2^{43.77}$

Table 2: Comparison of the probability of selecting a weak key (denoted by $\mathbb{P}$) and the costs of running the attack on a single instance (denoted by $\mathbf{c}$).

We conclude this section with a comparison table presenting the costs of the attacks described, alongside the original Prange’s ISD algorithm and the currently most efficient attack on BIKE, computed accordingly to *CryptographicEstimators*², when the parameters for the first and third levels of security are used.

²<https://github.com/Crypto-TII/CryptographicEstimators>

Security level	Attack 1	Attack 2	Prange's ISD	Best known attack
I	$2^{158} < \mathcal{C} < 2^{160}$	$\mathcal{C} \approx 2^{166}$	$\mathcal{C} \approx 2^{174}$	$\mathcal{C} \approx 2^{146}$
III	$2^{223} < \mathcal{C} < 2^{225}$	$\mathcal{C} \approx 2^{232}$	$\mathcal{C} \approx 2^{241}$	$\mathcal{C} \approx 2^{210}$

Table 3: Comparison of attack costs (denoted by $\mathcal{C}$) against BIKE.

7 Conclusions

The framework introduced in [12] has started a novel line of algebraic cryptanalysis for QC-MDPC code-based cryptosystems, highlighting possible vulnerabilities of BIKE and similar schemes. In this work we have presented two extensions of the state of the art.

- The first one is the introduction of a new necessary hypothesis for successful attacks, namely, the weak keys have to be pairs of coprime polynomials. Due to this, we were able to provide more accurate estimations of the computational cost of the key-recovery in [12].
- The second one is the introduction of a variant of a known attack, corresponding to a new larger set of weak keys that leads to an improvement of the attack cost.

Overall, the analysis of the computational costs shows how this framework is quite promising, as highlighted in particular by the comparison with the cost of Prange ISD. It is important to notice, however, that despite the advancements presented in this paper, the class of algebraic key-recovery attacks described does not currently break BIKE.

As extensions of this work, we plan to understand whether techniques such as enumeration, meet-in-the-middle search and representations can be borrowed from the ISD framework and applied to our algorithm, as well. Moreover, we plan to understand whether our approach can be used to produce codewords whose weight is moderately small (say, something in between the minimum distance and the Gilbert-Varshamov bound). Existence of such codewords would first provide a distinguisher for QC-MDPC codes. Moreover, from the knowledge of such codewords one may obtain the secret key, using techniques such as [16, 17].

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Appendix A BIKE

A.1 Setup

There are some global parameters used in BIKE that are chosen based on the desired security level λ to be achieved.

- The first parameter is r : a positive prime integer that determines the structure of the polynomial ring $\mathcal{R} := \frac{\mathbb{F}_2[x]}{(x^r-1)}$. r has to be chosen such that 2 is primitive modulo r , i.e. such that 2 is a generator of the multiplicative group of integers modulo r . Once that r is fixed, the code length becomes $n = 2r$;
- The second parameter is w : the row weight of the parity-check matrix. w has to be chosen such that $w/2$ is odd and $w \approx \sqrt{2r} = \sqrt{n}$. This parameter determines the public key space $\mathcal{H}_w := \{(h_0, h_1) \in \mathcal{R}^2 \mid |h_0| = |h_1| = w/2\}$;
- The third parameter is t : a positive integer such that $t \approx \sqrt{n}$. t represents the error weight and defines the error space $\mathcal{E}_t := \{(e_0, e_1) \in \mathcal{R}^2 \mid |e_0| + |e_1| = t\}$;
- The fourth parameter is l : a positive integer representing the shared secret size. In particular, once that l is fixed the message space $\mathcal{M} = \{0, 1\}^l$ and the shared key space $\mathcal{K} = \{0, 1\}^l$ can be defined.

Note that, since 2 is primitive modulo r , the polynomial $x^r - 1$ can be factorized over $\mathbb{F}_2[x]$ as

$$x^r - 1 = \prod_{d|r} \mathcal{Q}_d(x) = (x + 1) \mathcal{Q}_r(x),$$

where $\mathcal{Q}_r(x) = x^{r-1} + x^{r-2} + \dots + x + 1$ is the r -th cyclotomic polynomial. It is worth highlight that $\mathcal{Q}_r(x)$ is irreducible in $\mathbb{F}_2[x]$ because 2 is primitive modulo r . Consequently a polynomial in $\mathcal{R}$, different from $\mathcal{Q}_r(x)$, is irreducible if it has odd Hamming weight greater than 1.

Since BIKE uses the FO transform [18], in the Encapsulation and Decapsulation algorithms some hash functions are needed. BIKE uses three such functions with the following structure:

- $\mathbf{H} : \mathcal{M} \times \mathcal{M} \rightarrow \mathcal{E}_t$;
- $\mathbf{K} : \mathcal{M} \times \mathcal{R} \times \mathcal{M} \rightarrow \mathcal{K}$;
- $\mathbf{L} : \mathcal{R}^2 \rightarrow \mathcal{M}$.

A last element chosen in the Setup phase is the decoder `DECODER`, that takes in input an element $s \in \mathcal{R}$ and a private key $(h_0, h_1) \in \mathcal{H}_w$ and returns either $(e_0, e_1) \in \mathcal{R}^2$ such that $e_0 h_0 + e_1 h_1 = s$ or $\perp$.

A.2 Algorithm

The error-correcting code $\mathcal{C}$ used in BIKE is a (2,1)-QC-MDPC code of length $n = 2r$ and dimension $k = r$.

Let us now detail the KeyGeneration, Encapsulation and Decapsulation algorithms. Let $\pi_l : \mathcal{R} \rightarrow \mathcal{M}$ denote the function that give as output the first l bits of the input.

KeyGeneration

- **Input:** $\emptyset$.
 - **Output:** the private key $(h_0, h_1) \in \mathcal{H}_w$, $\mu, \sigma \in \mathcal{M}$, and the public key $h \in \mathcal{R}$.
1. Generate the private key $(h_0, h_1) \xleftarrow{\$} \mathcal{H}_w$;
 2. Compute the public key $h \leftarrow h_1 h_0^{-1}$;
 3. Compute $\mu \leftarrow \pi_l(h)$;
 4. Sample $\sigma \xleftarrow{\$} \mathcal{M}$.

Encapsulation

- **Input:** the public key $h \in \mathcal{R}$.
 - **Output:** the symmetric key $K \in \mathcal{K}$ and its encapsulation $c \in \mathcal{R} \times \mathcal{M}$.
1. Sample $m \xleftarrow{\$} \mathcal{M}$;
 2. Compute the error polynomials $(e_0, e_1) \leftarrow \mathbf{H}(m, \pi_l(h))$;
 3. Compute $c \leftarrow (e_0 + e_1 h, m \oplus \mathbf{L}(e_0, e_1))$;
 4. Compute $K \leftarrow \mathbf{K}(m, c)$.

Decapsulation

- **Input:** the private key $(h_0, h_1) \in \mathcal{H}_w$, $\mu, \sigma \in \mathcal{M}$ and the encapsulated key $c = (c_0, c_1) \in \mathcal{R} \times \mathcal{M}$.
 - **Output:** $K \in \mathcal{K}$.
1. Compute $e' \leftarrow \text{DECODER}(c_0 h_0, h_0, h_1)$;
 2. If $e' = \perp$ then $e' \leftarrow (0, 0)$;
 3. Compute $m' \leftarrow c_1 \oplus \mathbf{L}(e')$;
 4. If $e' = \mathbf{H}(m', \mu)$ then $K \leftarrow \mathbf{K}(m', c)$, else $K \leftarrow \mathbf{K}(\sigma, c)$.

Appendix B Confidence Intervals

A confidence interval (CI) is a range of values derived from a sample data that is likely to contain the true value of an unknown population parameter.

Let us now consider a random sample $(X_1, X_2, \dots, X_n)$ drawn from a population following a Bernoulli distribution with parameter p . In this context, each X_i is a binary random variable that takes the value 1 with probability p , and 0 with probability $1 - p$. Our goal is to determine the confidence interval for the parameter p , denoted by $\text{CI}_p(1 - \alpha)$. This confidence interval will give us a range of plausible values for p with a specified confidence level of $1 - \alpha$ (or equivalently with a significance level of α).

Definition 3 Let $(X_1, X_2, \dots, X_n)$ be a random sample from a given distribution. The *Sample Mean Estimator*, denoted by $\bar{X}_n$, is defined as

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i.$$

Our aim is to find the value of a that satisfies the equation

$$\mathbb{P}(|\bar{X}_n - p| \leq a) = 1 - \alpha,$$

where p denotes the true value of the target population parameter.

In other words, we are seeking the value of a that ensures, with a probability of $1 - \alpha$, that the difference between the sample mean $\bar{X}_n$ and the true probability p falls within the range $[-a, a]$.

The equation can be further transformed into

$$\mathbb{P}\left(-\frac{a}{\sqrt{\frac{p(1-p)}{n}}} \leq \frac{\bar{X}_n - p}{\sqrt{\frac{p(1-p)}{n}}} \leq \frac{a}{\sqrt{\frac{p(1-p)}{n}}}\right) = 1 - \alpha.$$

To arrive to a more precise form of the confidence interval formula, let us refresh some useful details.

Since $X_1, \dots, X_n$ are independent identically distributed Bernoulli random variables of parameter p , then $\sum_{i=1}^n X_i$, as well as $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, is a binomial random variable with parameters n and p .

Since $\text{Var}(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{p(1-p)}{n}$, for the *Central Limit Theorem*³

$$\frac{\bar{X}_n - p}{\sqrt{\frac{p(1-p)}{n}}} \sim \mathcal{N}(0, 1).$$

This connection between the binomial distribution and the standard normal distribution allows us to express the values $-\frac{a}{\sqrt{\frac{p(1-p)}{n}}}$ and $\frac{a}{\sqrt{\frac{p(1-p)}{n}}}$ as the quantiles $z_{\frac{\alpha}{2}}$ and $z_{1-\frac{\alpha}{2}}$, respectively.

Note that, since we are working with a two tailed CI, we need to split the significance level α in half to find the relevant z-values from the standard normal distribution table.

We can write

$$\text{CI}_p(1 - \alpha) = \left[\bar{X}_n + z_{\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}}; \bar{X}_n + z_{1-\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}} \right].$$

³for a detailed explanation of the Central Limit Theorem refer e.g. to [19] or [20]

Since we are estimating p itself, we can replace $\text{Var}(X_i) = p(1 - p)$ with its unbiased estimator.

Definition 4 Let $(X_1, X_2, \dots, X_n)$ be a random sample from a given distribution. The *Sample Variance Estimator* S_n^2 is defined as

$$S_n^2 := \frac{\sum_{i=1}^n (X_i - \bar{X}_n)^2}{n - 1}.$$

This procedure yields the following formulation of the searched confidence interval

$$\text{CI}_p(1 - \alpha) = \left[\bar{X}_n + z_{\frac{\alpha}{2}} \frac{S_n}{\sqrt{n}}; \bar{X}_n + z_{1-\frac{\alpha}{2}} \frac{S_n}{\sqrt{n}} \right].$$

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